Question	Answer	Mark	Mark scheme	Additional guidance
22	0.5	P1	derive an algebraic expression for the area of A eg $\frac{1}{8}\pi\,[\,(5x-1)^2-(3x-1)^2\,]$	
		P1	expand and simplify for either area A or area B eg $\frac{1}{8}\pi\,(16x^2-4x)$ **or** $\pi(x^2-2x+1)$	
		P1	(dep P2) equate and rearrange into a quadratic eqn of the form $ax^2 + bx + c = 0$ eg $2x^2 + 3x - 2 = 0$	
		P1	(dep P3) factorise eg $(2x - 1)(x + 2) = 0$ **or** use of formula eg $\frac{-3\pm\sqrt{3^2-4\times2\times-2}}{2\times2}$	
		A1	oe	Accept only the single value of 0.5 oe but award 0 marks for a correct answer with no supportive working